Subpixel Melt Index in the Antarctic Peninsula Using Spatially Constrained Linear Unmixing From Time Series Satellite Passive Microwave Images

Dan Tian and Lei Wang, *Member, IEEE*

***Abstract*— The inevitable coarse resolutions (∼25 km) of satellite passive microwave images introduce large uncertainty to surface melt estimation on Antarctic ice margins. Our test showed that the melt index (MI) of the Austral year 2012–2013 in the Antarctic Peninsula calculated from a high-resolution product of Calibrated Enhanced-Resolution Passive Microwave Daily EASE-Grid 2 Brightness Temperature (CETB) was 33% lower than the original low-resolution Special Sensor Microwave Imager/Sounder (SSMIS) images. Therefore, to allow for fractional melt estimation, a subpixel mapping method was adopted in this research to improve the accuracy and reliability of surface melt measurement from passive microwave images. This method uses the least-squares mixture analysis (LSMA) on the time series of daily passive microwave images by taking advantage of their high temporal resolution. The endmembers of the unmixing calculation were collected under the constraint of Voronoi polygons. The fractional MI was calculated by multiplying its area with melt fraction. Using the CETB images as reference, we found that the overestimation of surface melt using the Boolean classification was corrected by the LSMA method. The root-mean-square difference (RMSD) of the MI derived from SSMIS compared with the CETB data was reduced from 0.0104 to 0.0083 × 10^6 day·km^2. A log-linear relationship was found between melt fraction and elevation, which showed that melt fraction is inversely correlated with elevation, and topography is a dominant factor for its variation in low elevations. Such a subpixel unmixing analysis on the passive microwave images can improve the accuracy and reliability of surface melt mapping.**

***Index Terms*— Antarctic surface melt, least-squares mixture analysis (LSMA), time series analysis of passive microwave data.**

I. Introduction

MAPPING surface melt on Antarctic Ice Shelves is of great significance for studying glacier dynamics and regional and global climate changes. Melting snow has a low visible and near-infrared albedo, absorbing up to four times as much solar radiation as dry snow [1]. In addition, percolating melt water accelerates iceberg calving and causes ice mass loss and ice front retreat [2], [3]. Antarctica Ice Shelves are sensitive to the warming temperature. As shown in [4], surface melt intensity is related to the locally accumulative number of days above the melting temperature. Continuous and intense surface melt contributes to enlarged ice crevasses [2], accelerated glacier flow [5], and disintegration of ice tongues [6]. Monitoring surface melt phenomena is a key to understanding regional and global climate changes [7]–[9].

Owing to higher summer air temperature, the surface melt intensity in the Antarctic Peninsula accounted for more than 50% of the total in Antarctica for the past several decades [10], [11]. Extensive and long-time surface melt was known to accelerate ice shelve retreat, causing catastrophic events such as the collapse of Larsen B Ice Shelf in the austral summer of 2001/2002 [6], which was followed by significant accelerations of land-based glaciers [12], [13].

Antarctica has the highest winds, lowest temperatures, and highest average elevation on earth. Severe weather, poor accessibility, and polar nights give rise to serious challenges to traditional field observations. The surface melting in the polar regions has been studied using brightness temperature (T_b) measured by spaceborne passive microwave radiometers [4], [8], [9], [14]–[26] and using backscattering (σ_0) measured by radar scatterometers [3], [10], [27] and synthetic aperture radars (SAR) [7], [28], [29]. Compared with radar scatterometers or passive microwave radiometers, SAR has higher spatial resolution and can therefore detect more details [30]. However, it also has several drawbacks such as lower temporal resolution, significant geometric distortions, and topographic effects [30], [31]. Radar scatterometers have greater spatial and temporal resolution and are also more sensitive to the liquid water content among snow related to passive microwave products, but the observational time series length is limited [10]. Since 1978, long-term passive microwave time series has been acquired by the Scanning Multichannel Microwave Radiometer (SMMR) onboard the Nimbus-7 satellite, and a series of Special Sensor Microwave Imager (SSM/I) and Special Sensor Microwave Imager/Sounder (SSMIS) sensors from the Defense Meteorological Satellite Program (DMSP) [32]. The data have been widely applied to map glacier and ice sheet surface melting for Greenland [17], [22]–[26] and Antarctica [4], [14]–[18]. The over 40-year time series data offer an opportunity to study interannual variations in surface melting

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The authors are with the Department of Geography and Anthropology, Louisiana State University, Baton Rouge, LA 70810 USA (e-mail: dtian2@lsu.edu; leiwang@lsu.edu).

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and explore its relationship with climate change in the modern human society. Such works, however, are limited by the coarse spatial resolution of the passive microwave products.

A large instantaneous field of view (IFOV) is required for passive microwave sensors on satellites to achieve sufficient signal-to-noise ratio at 700–800-km altitude. The IFOV of SSMIS at the 19-GHz channel, defined as the 3-dB footprint size, projects on the earth surface as ellipses of about 42 km $\times$ 70 km. To derive a gridded product, for example, on the Equal-Area Scalable Earth (EASE) grid, the signal is resampled to the SSMIS EASE grids with a nominal cell size of 25 km by Backus–Gilbert interpolation using observations from a single orbit [32], [33]. The coefficients are tuned to reduce noise, producing optimally filtered data, which means T_b could have been measured if the sensor is aligned with the grid center [32], [33]. However, uncertainty due to the large pixel size exists in the melt maps derived at the 25-km scale and can be large enough to distort our knowledge of climate changes derived from the melt maps. Consistent negative trends of surface melt over the Antarctic continent were reported by different research groups using the satellite passive microwave data [8], [16], [17], [34]. These reported trends, however, were contradictory to the warming climate records in Antarctica [35] and increasing sea levels caused by ice sheet melt [36]. The uncertainty due to large pixel size of surface melt records from the passive microwave data could contribute to such inconsistency between the surface melt trend in Antarctica and other climatological reports. The Peninsula is one of the most studied areas of Antarctica because of its low latitude and the mildest climate in Antarctica. This area is sensitive to the warming temperature because of the flat terrain on the Larsen Ice Shelf, which occupies one-third of the area. The elevation changes from the sea level to 2800 m in a short transit. The complex terrain also makes the large pixel size images difficult to describe the surface changes.

The work by De Freitas *et al.* [27] suggested that the Boolean classification of melt and nonmelt status of the large SSMIS pixels would overestimate melt extents. They proposed an unmixing method using four channels (19H, 19V, 37H, and 37V) of SSMIS to unmix three endmembers: wet snow, dry snow, and rock outcrops. Their method, however, is limited by the number of channels (p) to unmixing up to $p - 1$ endmembers [37]. The high similarity among the four channels in their responses to melting and nonmelting status compromises the unmixing model premise. T_b of dry snow can range from 175 to 235 K because of the large variation in the emissivity, which is strongly dependent on the grain size [4]. Therefore, it is difficult to determine pure dry endmembers based only on the absolute T_b. In this research, we propose a new unmixing algorithm that: 1) treats the time series T_b data as hyperspectral images and 2) identifies endmembers by searching spatial distances. The objective of this article is twofold: 1) to quantify the overestimation bias of the existing SSMIS surface melt maps and 2) to verify that our new unmixing method can reduce the uncertainty and correct the bias of melt map production from passive microwave remote sensing data.

II. Data and Study Area

As the predecessor of the SSM/I-SSMIS sensors, the SMMR sensor measured T_b from October 1978 to August 1987. The SSM/I-SSMIS sensors continued the measurement from September 1987 up to now. The SMMR sensor had ten channels, which are both horizontal and vertical polarizations for 6, 10, 18, 21, and 37 GHz. The SSM/I sensors had seven channels, including both polarizations for 19, 37, and 85 GHz, as well as the vertical polarization for 22 GHz. The SSMIS sensors provide data from 24 channels between the frequencies of 19 and 183 GHz, including the five channels at 19, 22, and 37 GHz, which are the same as the SSM/I channel configuration.

We used the SSMIS daily EASE-grid brightness temperature data from the 19H channel at the satellite ascending nodes from July 1, 2012, to June 30, 2013, for this study. This low-frequency channel was chosen because the magnitude of the transition of T_b from winter to summer is smaller at higher frequencies because of stronger scattering during winter [16]. According to Abdalati and Steffen [25] and Liu *et al.* [16], at the same frequency, the conversion from dry snow to wet snow leads to a greater increase in T_b of the horizontal polarization than that of the vertical polarization. The local time of ascending nodes is around 6:00 P.M., when T_b reaches its maximum [38], which helps eliminate the disturbances of refreezing during night.

For validation of the subpixel fraction derived from the unmixing algorithm, we used the Calibrated Enhanced-Resolution Passive Microwave Daily EASE-Grid 2 Brightness Temperature (CETB) Earth System Data Record (ESDR) produced from the MEaSUREs program funded by NASA, which used the radiometer version of the Scatterometer Image Reconstruction (rSIR) algorithm [39], [40] to reprocess and fuse the complete data records of SMMR, SSM/I-SSMIS, and AMSR-E in a single grid passive microwave ESDR with improved resolution. The rSIR algorithm takes advantage of overlapping and oversampling satellite IFOVs to reconstruct the high-resolution image [41]. Measurements from satellite passes within a small time window are combined by a weight based on the measurement response function (MRF), which is identified by the antenna gain pattern, the scan geometry, and the integration period [42].

Since the sampled MRF is large, sparse, and easy to be overdetermined or underdetermined, iterative methods are applied. An enhanced resolution is accompanied by increasing noise. The rSIR algorithm limits the noise by truncating its iteration [39], [40]. CETB provides morning/evening local time-of-day (LTOD) pass for the Southern images. We used the evening LTOD pass 6.25-km rSIR temperature images from the F-16 SSMIS sensor covering the same region and time as the DMSP SSMIS data we used. The evening LTOD pass is generated by combining measurements between 12:00 P.M. and 12:00 A.M. Compared with the single-pass SSMIS data at around 6:00 P.M., the evening LTOD pass CETB data might be affected by the refreezing at night. Each original SSMIS pixel encompasses 16 CETB pixels. We used the melt fraction derived from each set of 16 CETB pixels to verify

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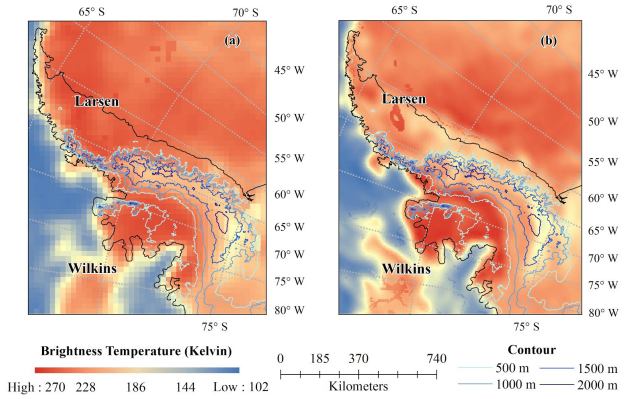


Fig. 1. Study area of the Antarctic Peninsula. (a) SSMIS image and (b) CETB image of December 31, 2012, for the 19-GHz horizontal polarization.

the melt fraction calculated from the least-squares mixture analysis (LSMA) described in Section III.

Fig. 1 shows the boundary, coastline, and contour lines of the study area overlaid with the SSMIS image (a) and the CETB image (b) of December 31, 2012, from 19H channel. The study area measures as 1150 km × 1400 km (a grid of 46 columns and 56 rows), with 584 375 km^2 of land surface area, including ice shelves at both sides of the Peninsula.

III. Methods

A. Melt Detection

On ice sheets, ice shelves, and sea ice surrounding the Antarctica continent, passive microwave sensors detect naturally emitted microwave energy from both dry and melting snow within their field of view [8], [17], [18], [43]–[45]. Passive microwave is in the form of T_b calibrated based on the recorded microwave energy. The relationship between the calculated T_b and the near-surface physical temperature (T_s) can be expressed by first-order Rayleigh–Jeans approximation [46] in the microwave spectra

$$T_b = \varepsilon \cdot T_s \tag{1}$$

where ε is the emissivity of snow near the surface. The emissivity of dry snow is dependent on the firn grain size and snow depth. The microwave electromagnetic radiation generated from the deeper layers in snow packs is scattered by ice grains, which reduces the emissivity [4]. Increased snow depth will cause more reduction in emissivity. Therefore, dry snow emissivity at the 1.5-cm wavelength (corresponding to the 19-GHz radiometer frequency) varies from 0.65 to 0.85, while the wet snow emissivity at this wavelength is about 0.9. Assuming the temperature at the melting point of ice is 273 K, the theoretical T_b of wet snow is 246 K, which is about 15–70 K above T_b of dry snow [4]. Note that for snow overlying an ice cap or sheet, the difference between the surface temperature and the temperature of the main emitting layer, which can be situated a few ten centimeters below the surface of dry snow at 19 GHz, is of the order of one to a few Kelvin degrees at maximum. Therefore, the effect of using the surface instead of an emitting layer temperature on T_b is small compared with the T_b change caused by melt onset. When dry snow packs start to melt, T_b will rise drastically even if the physical temperature does not change. Similarly, T_b will drop back to the premelting status when the snowpack refreezes. Microwave sensors can detect the increase and decrease in T_b when snow begins to melt and refreeze, respectively [16]. Such T_b changes, rather than physical temperature fluctuations, can be used to track the melt and refreeze events during an austral summer from satellite observations.

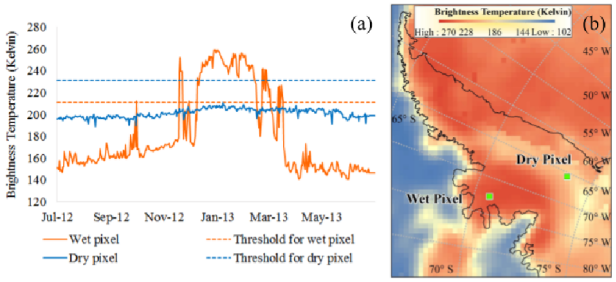


Fig. 2. (a) Annual variations in T_b for typical wet and dry pixels under the 19-GHz horizontal channel of the SSMIS data from July 1, 2012, to June 30, 2013, as well as their adaptive melt detection thresholds according to Zwally and Fiegles [4]. (b) Locations of the two pixels.

For better generalizability, the algorithm for melt detection proposed by Zwally and Fiegles [4] was chosen rather than sophisticated edge detection methods such as Liu *et al.* [47]. In [4], a local mean T_b plus 30 K (noted as ZF+30) was used as the melt detection threshold for each pixel. This algorithm takes the spatial variation in dry snow emissivity into consideration and allows the threshold values to be adaptive to the local conditions. Fig. 2 compares the time series of a wet pixel (pixel that experienced melt in the year) and a dry pixel (pixel that did not melt in the year), along with the Zwally and Fiegles threshold (ZF+30) lines for melt detection in both time series. Thus, the melt detection algorithm assigns 73 melting days to the wet pixel and 0 melting days to the dry pixel.

We used melt index (MI) as the variable to compare the melting status between data set with different pixel sizes. MI (in day·km^2) is the melt duration times melt extents for a certain area (e.g., the entire Antarctica or the Antarctic Peninsula) for the entire season [16]. MI is a convenient variable to study the interannual trends of surface melt in Antarctica and Greenland and an indicator for climate change [8], [16], [43], [50]. The formula of the MI is

$$\mathrm{MI} = \sum_{i=1}^{s} \mathrm{md}_i * A \tag{2}$$

where A is the area size of one pixel, which is a constant for all pixels, s is the total number of pixels, and md_i is the number of melt days (melting duration) within a year for pixel i. With a grid size of 25 km × 25 km, the MI of the wet pixel in Fig. 2 is computed as: 73 day × 625 km^2 = 45 625 day·km^2.

B. LSMA of Partial Melt Data

Fig. 3 illustrates an example of time series for a full melt pixel and a partial melt pixel as well as their melt detection thresholds. The MI for the full melt pixel is 55 625 day·km^2 (89 days with T_b greater than its threshold, which is around

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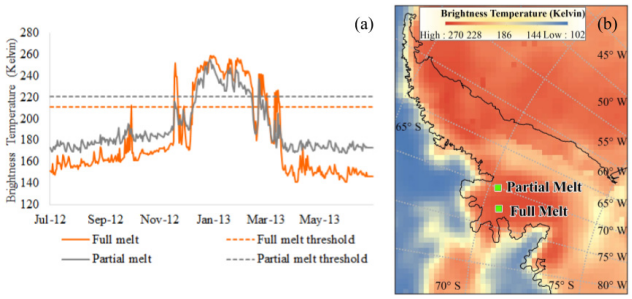


Fig. 3. (a) Annual variations in T_b for typical full melt and partial melt pixels under the 19-GHz horizontal channel of the SSMIS data from July 1, 2012, to June 30, 2013. (b) Locations of the two pixels.

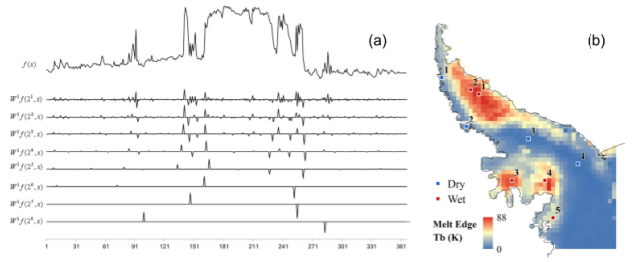


Fig. 4. (a) Daily T_b time series of endmember melt (wet) #3 and its wavelet transformation local maxima [47] at all scales from day 1 (July 1, 2012) to day 365 (June 30, 2013). Particularly, $W^1 f(2^4, x)$ at the fourth scale was used to classify the edges to large (GL) and small (GS) groups. (b) Locations and labels of five melt (wet) endmembers from high-value areas and four nonmelt (dry) endmembers from low-value areas. The warm colors are high values of mean edge magnitude and the cool colors are low values.

210 K), and the partial melt pixel is 40 000 day·km^2 (64 days with T_b greater than its threshold, around 220 K). The full melt pixel and the partial melt pixel have similarly shaped time series plots, indicating the synchronization of the two pixels in melting occurrence due to their closeness in distance. However, because the pixel, which covers an area of 625 km^2, might contain a mixture of melting and nonmelting subregions, T_b at beginning and end of the melt status does not exceed the threshold, leading to a shorter melting duration and a smaller MI. The dilemma arises for melt-day calculation: do we need to decrease the threshold to assign more melting days for the partial melt pixel or keep the threshold but decrease the melting days? From the time series view, it seems the melt duration was underestimated in the illustrated partial melt pixel. To correct the underestimation problem, a lower threshold should be proposed, which requires the knowledge of each partial melt pixel. Even if the new threshold is assigned, from the spatial point of view, the Boolean assignment of melt overestimates the MI because only a fraction of the 25 km $\times$ 25 km pixel experienced melting. This ambiguity can be solved by calculating abundance fraction of melting and nonmelting subregions within each partial melt pixel using the LSMA. Explicitly, let $\boldsymbol{t}$ be a $l \times 1$ column vector representing one-year T_b time series for a pixel, with $l = 365$ (or 366 in a leap year). T_b of a partial melting pixel in a time series vector $\boldsymbol{t_p}$ is a linear combination of the T_b time series vector of melt endmember $\boldsymbol{t_m}$ and nonmelt endmember $\boldsymbol{t_n}$, as expressed by

$$\boldsymbol{t_p} = \alpha_m \boldsymbol{t_m} + \alpha_n \boldsymbol{t_n} + \boldsymbol{\varepsilon} \tag{3}$$

where α_m is the fraction of the melting area; α_n is the fraction of the nonmelting area; and ε is the error term in the form of an $l \times 1$ column vector. The least-square solution, which minimizes the error term, under the constraints of unity and positiveness [48], is used to solve the equation. Once the melt fraction (α_m) is calculated, the melt area can be updated by multiplying the total area of an SSMIS pixel and the fraction number. For example, if a pixel is 50% melt, the total melt area from that pixel is 25 km $\times$ 25 km $\times$ 50% = 312.5 km^2 instead of 625 km^2.

C. Endmember Selection

In (3), endmembers of melt and nonmelt time series should be provided to obtain the solution of melt and nonmelt fraction for each partial melt pixel. Endmembers are defined as pure pixels, which represent the information of a homogeneous phenomenon such as melting or nonmelting. $\boldsymbol{t_n}$ is the time series of a nonmelt pixel in a year. $\boldsymbol{t_m}$ is the time series of a full melt pixel. It is relatively easy to find the nonmelt pixels using the ZF+30 criterion. Any pixel whose T_b never exceeds this threshold belongs to nonmelt pixels. However, as shown in Fig. 3, both the full melt and partial melt pixels exceed the ZF+30 criterion. Therefore, the ZF+30 criterion cannot separate full melt and partial melt pixels. However, as can be observed from Fig. 3, the melt edges (the sharp rise and fall in the T_b curve caused by the onset and end of melting) of the full melt pixel is greater than those of the partial melt pixel because the mixture of melt and nonmelt signal tends to reduce the magnitude of these edges.

It is not difficult to run a wavelet decomposition and use a bimodal classification method to differentiate melt edges from nonmelt edges at the fourth wavelet scale [47]. Fig. 4(a) shows the time series of melt endmember #3 in Fig. 4(b) and its wavelet transformation local maxima used to track melt occurrences [47]. The wavelet transformation filters the noise component and preserves stronger edges at higher scales as indicators of melt events. The bimodal classification sorts the edges by their magnitude and groups them into two classes—large (GL) and small (GS). The goal is to minimize the in-group variance and maximize the between-group variance [47]. For a wet pixel, GL contains the edges caused by melt signals, while GS contains the edges caused by temperature fluctuations. For nonmelt pixels, both groups are nonmelt edges, which are low in magnitude. Therefore, the mean magnitude of the GL of a nonmelt pixel is much lower than that of a melt pixel. The mean magnitude of the GL is used as an indicator of pixel purity: the larger the mean GL edge magnitude, the more likely the pixel to be a pure melt pixel. Similarly, a small mean GL edge magnitude indicates that the pixel is likely to be a nonmelt endmember. Fig. 4(b) shows the GL edge magnitude map in the study area and the locations of melt and nonmelt endmembers. Indeed, all the red-colored pixels could be considered as areas of full melt. We selected the most representative ones as endmembers, which are the pixels with the largest GL value within a wet area. The blue-colored area contains nonmelt endmembers, and beige colors represent mixed pixels. We selected these endmembers to run

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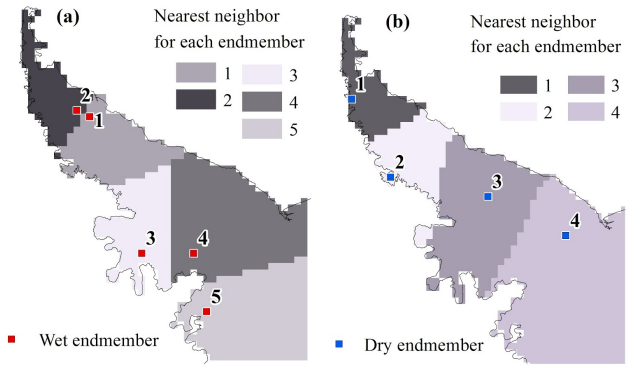


Fig. 5. Spatially constrained endmember selection defined by Voronoi polygons. (a) Nearest neighbors for melt (wet) endmembers. (b) Nearest neighbors for non-melt (dry) endmembers.

the LSMA for melt fraction calculation within each 25-km SSMIS pixel. The melt edges are clearly much stronger than nonmelt edges, not only in the magnitude but also in the temporal continuity.

The endmembers are defined locally because the shape of the T_b time series changes from region to region. Each melt event is bounded by certain terrain boundaries such as ridges of high mountains on ice sheets. On the other hand, the melt events on ice shelves could extend far because of the flat topography. Endmembers must be representative and inclusive. Due to the spatial stationarity of the microwave observation among the dry snow areas, the time series shapes of the adjacent dry snow pixels are like each other. Therefore, we selected endmembers from the centroid of a dry/wet snow area to ensure representativity. To be inclusive, endmembers need to be diverse, which is ruled by the distance from each other, because according to the first law of Geography [49], data similarity will decline with the increasing distance. The four nonmelt endmembers in Fig. 5 were selected for each subregion of the study area and segregated from each other. The two melt endmembers #1 and #2 represent the mixture condition where both the grounding lines and coast lines overlap. In these areas, elevation can change drastically within several kilometers, and therefore even a dry pixel contains some portion of information from low-lying coastal regions. Nonmelt endmembers #3 and #4 were selected from the inner part of the ice sheets. Therefore, their T_b represents only high elevation areas.

For each mixed pixel, the melt and nonmelt endmembers were picked from a tessellation of Voronoi polygons defined by the locations of endmembers (Fig. 5). Each Voronoi polygon defines the nearest neighbor area to the endmember candidates. Mixed pixels located within the polygon defined by an endmember are unmixed from it. The tessellation map of Fig. 5(a) is for melt endmember selection. The tessellation map of Fig. 5(b) is for nonmelt endmember selection. The nearest neighbor zones assign one melt endmember based on Fig. 5(a) and one nonmelt endmember based on Fig. 5(b) to the partial melting pixels. With the melt and nonmelt endmembers, the fractions of the melt and nonmelt area in each partial melting pixel are computed using (3).

TABLE I
COMPARISON OF MI FROM DIFFERENT DATA SETS AND THEIR RMSD REFERENCED TO CETB DATA

Melt index (10^6 day·km^2)	SSMIS	LSMA	CETB
Study area total	11.365	6.478	7.565
RMSD	0.0104	0.0083	N/A

D. Accuracy Assessment

We used the 6.25-km CETB product to verify the unmixing model. The product is created by fusing multiple passive microwave observations from overlapping orbits of the same sensor and/or different sensors to reconstruct the super-resolution T_b data. In terms of the spatial resolution, this product is comparable with another high-resolution product from the QUIKSCAT satellite SeaWinds Ku-band radar image using a similar high-resolution reconstruction algorithm to enhance its spatial resolution up to 2.25 km. In fact, the super-resolution QUIKSCAT product was used for accuracy assessment in many snow melt mapping projects [27], [50]. However, the problem of QUIKSCAT data is the short data collection campaign. The CETB ESDR program creates super-resolution passive microwave images spanning much longer time than QUIKSCAT, and because it is a product of radiometers rather than scatterometers, it is superior to QUIKSCAT images for verifying the unmixing algorithm on passive microwave images.

The comparison between the model output and the high-resolution data uses the MI derived from the time series of passive microwave data. Each 25 km × 25 km SSMIS pixel contains 16 high-resolution CETB pixels (6.25 km × 6.25 km). The total MI of the 16 CETB pixels nested within one SSMIS pixel is compared with the unmixed SSMIS pixel MI from LSMA. The sum of MI for the entire study area and the root-mean-square difference (RMSD) from pixel-wise comparison of all the land pixels are reported in Table I. The RMSD is calculated as follows, where $\hat{MI}$ is the estimated MI and MI is the reference MI

$$\mathrm{RMSD} = \sqrt{\frac{\sum_{i=0}^{N} (\hat{MI}_i - \mathrm{MI}_i)}{N}}. \tag{4}$$

IV. RESULTS AND DISCUSSION

A. Linear Mixture Analysis and Accuracy Assessment

Table I summarizes the total MI of the Antarctic Peninsula and pixel-wise statistical values calculated from the three data sets: 1) Boolean SSMIS; 2) LSMA SSMIS; and (3) high-resolution CETB images. The Boolean MI of the SSMIS data is about 50% larger than the CETB MI, showing that there is a significant bias of overestimation of the reported surface melt using the original SSMIS data. This might be attributed to the fact that the ground IFOV (70 km ∗ 42 km) of SSMIS is about twice as large as the grid size (25 km) of its image product. Therefore, the bias caused by the topographic complexity in

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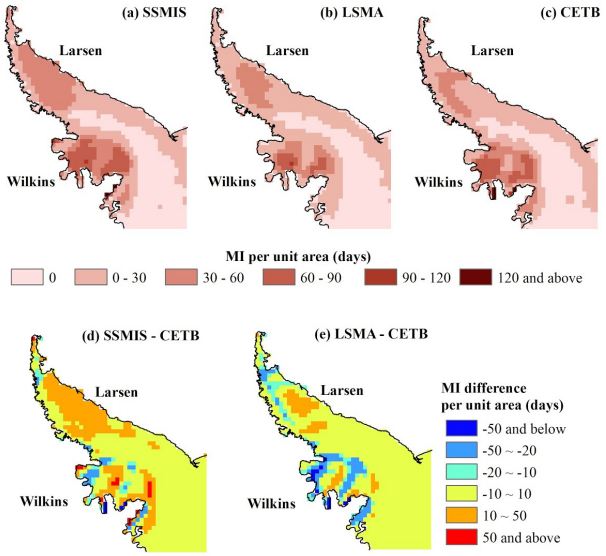


Fig. 6. Comparing MI of (a) Boolean SSMIS, (b) LSMA, (c) CETB, (d) SSMIS–CETB, and (e) LSMA–CETB result.

the study area is greater than the expected. As a comparison, the MI of LSMA is slightly (14%) smaller than the CETB MI.

The lower RMSD of the LSMA result indicates that the LSMA MI improved the confidence and accuracy over the Boolean MI from SSMIS images. The 14% lower MI of the LSMA result is consistent with previous works. Overestimation also exists in the Boolean melt calculation from the high-resolution CETB data as shown in Section IV-D. Therefore, the 14% lower MI from the LSMA of SSMIS images should be a more accurate representation of the real MI than the one from the original SSMIS images. The LSMA MI was satisfactory given the large bias of the Boolean melt products. Fig. 6 shows the maps of the Boolean SSMIS MI [Fig. 6(a)], the SSMIS LSMA MI [Fig. 6(b)], and CETB MI [Fig. 6(c)] for each land pixel, as well as the differences between SSMIS and CETB MI [Fig. 6(d)] and between LSMA and CETB MI [Fig. 6(e)]. Compared with the CETB MI map [Fig. 6(c)], the overestimation of the SSMIS MI mainly occurs on the Larsen Ice Shelf and the Wilkins Ice Shelf. The LSMA method overestimates the MI on the Larsen Ice Shelf but underestimates it on the Wilkins Ice Shelf. The overall pattern of the LSMA MI map matches the CETB MI map better than the original SSMIS MI, indicating a better accuracy of the estimated MI from the LSMA method.

B. Impact of Pixel Size and Endmembers Purity on LSMA

The LSMA model assumes that the wet and dry pixels [Fig. 4(b)] selected from the study area are pure endmembers. However, with the large pixel size of the 25-km SSMIS images, it is difficult to locate fully melted pixels. Therefore, we must acknowledge that the endmembers of wet pixels could only be approximations and uncertainty could be brought to the LSMA model by the impurity of endmembers. Strictly speaking, it may be impossible to find real pure endmembers to implement the unmixing method at such a scale as satellite radiometer mapping. It is interesting to test, however,

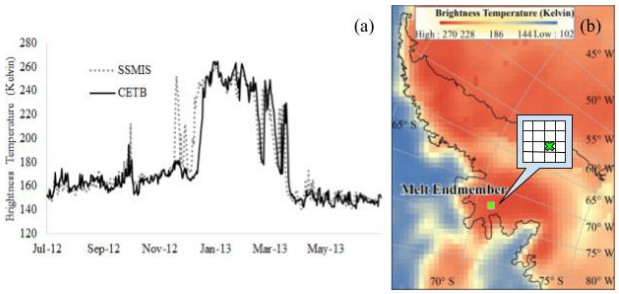


Fig. 7. (a) Comparison between the CETB pixel and the SSMIS pixel as endmembers and (b) location of the endmembers. The zoomed-in box shows the location of CETB endmember within the SSMIS endmember pixel.

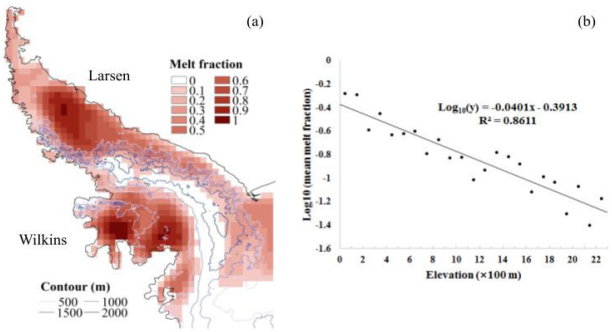


Fig. 8. (a) Melt fraction map calculated from LSMA and (b) scatterplot of between the melt fraction and mean elevation of the pixel.

if some relaxation on the pure endmember selection criterion is acceptable and if the LSMA model can be improved using the endmembers derived from the high-resolution product rather than from the same SSMIS scene.

Theoretically, high-resolution images should contain more pure pixels for unmixing. To test how much the impurity of endmembers have impacted the unmixing results, we choose endmembers from the high-resolution CETB images to implement the LSMA unmixing calculation. The CETB pixels were selected from the center location of the SSMIS endmember pixels. As shown in Fig. 7, the time series of the CETB endmember has fewer peaks during the melting season. In addition, the onset of the long melt event (from December of 2012 to late March 2013) is later than the SSMIS data. During the melting season, the CETB T_b values are slightly greater than SSMIS. Then, the LSMA is performed using the endmembers that are supposed to be "purer" because the resolution of the CETB pixels is better. However, the results show that the RMSD of unmixing the SSMIS images using the CETB endmembers is 0.0088 (10^6 day·km^2), which is very close to but 6.02% higher than that using the SSMIS endmembers. Therefore, although the endmember impurity could cause uncertainty for unmixing, the endmembers obtained from the high-resolution image do not improve the LSMA model. On the other hand, the high similarity between the unmixing results from SSMIS endmembers and the CETB endmembers indicates that the pixel resolution and impurity of endmembers do not add much uncertainty to the LSMA model.

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C. Relationship With Elevation

The spatial pattern of the melt fraction map in Fig. 8(a) shows a high resemblance with the topographic maps published for the Antarctic Peninsula. Therefore, we suspect the elevation may play an important role in the partial melting problem. We obtained a 1-km elevation model of Antarctica [51] to analyze its correlation with the melt fraction map. In the scatterplot [Fig. 8(b)], each elevation class is defined by a 100-m interval from the lowest to the highest and named by the integer number rounded up from the 100th of their elevation values. In each elevation class, the mean melt fraction is calculated to make the scatterplot with the elevation class. Because of the nonlinearity of the original scatterplot, a logarithm is taken on the mean melt fraction values. The melt fraction is almost 100% on the ice shelves near the coast and goes down sharply with the increasing elevation toward the mountains. The scatterplot shows a clear inverse log-linear relationship ($p < 0.0001$) between the elevation class and the mean melt fraction. It suggests that in the study area, the partial melting problem in the Antarctic Peninsula is mainly caused by the elevation change. The logarithm relationship between elevation and melt fraction indicates that the elevation has a major impact on melt fraction over low-lying and flat ice shelves, and that when the elevation is higher (e.g., 1000 m), the relationship between elevation and melt fraction is weaker. The melt fraction increases sharply along the slope facing the Bellingshausen Sea from zero to near full melting in a short distance. The side facing the Weddell Sea does not have such a strong trend. The difference could be caused by the effect of different temperature lapse rates and blocking effects of the mountains to clouds and precipitation.

D. Unmixing the High-Resolution CETB Images

From the above discussions, we have proved the ability of LSMA to correct the bias introduced by the Boolean MI. It is interesting to test how much difference in the MI is derived from high-resolution CETB images if the Boolean calculation is replaced by the LSMA model. The purpose of the test is to validate the application of LSMA to higher resolution image products. Therefore, we apply the LSMA to the high-resolution CETB images and compared the result with the LSMA MI from the SSMIS data.

Fig. 9(a) shows the LSMA MI map of the CETB data using the spatially constrained endmembers. Compared with the LSMA MI map from SSMIS images [Fig. 9(b)], the resolution and spatial structure is much clearer. The overall LSMA MI of the CETB data is 4.417 (10^6 day·km^2), which is about 41.6% lower than its Boolean MI. This difference is consistent with the comparison between the Boolean MI and LSMA MI of SSMIS images, where the LSMA MI is 43% lower. The overestimation bias of the Boolean calculation applied to CETB images was therefore corrected by the LSMA method. We used the high-resolution product of CETB images as a reference and confirmed such treatment improved the accuracy of the MI estimation. Although owing to the unavailability of a higher resolution product than the CETB data to verify its LSMA result, the LSMA CETB MI is a better representation of surface melt detected from passive microwave images than the Boolean MI maps.

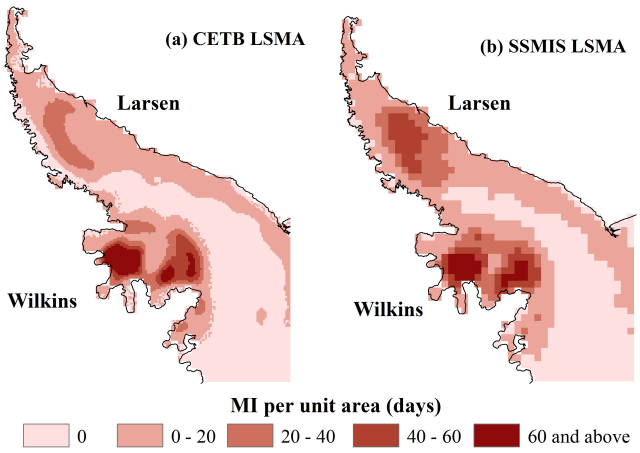


Fig. 9. (a) LSMA MI map from CETB images. (b) LSMA MI map from SSMIS images.

E. Data Quality Issue and Other Opportunities to Use LSMA

The use of SSMIS in this article is motivated by its consistency over a long period covered by its ancestors SSM/I and SMMR. However, the radiometer has a nominal ground IFOV of 70 km × 42 km at the 19-GHz frequency. Therefore, the actual spatial resolution of SSMIS images is lower than their 25 km × 25 km pixel size. The difficulty in finding good endmembers for LSMA is increased by the low resolution because the endmembers must be defined as "pure" pixels with homogeneous surface melting patterns. The enhanced resolution product CETB suffers from the same drawback of large ground IFOV, although using the overlapping tracks is an effective way to increase the spatial resolution.

An alternative radiometer, such as AMSR-E and AMSR2 with the ground IFOV of 27 km × 16 km and 22 km × 14 km for the 19-GHz channels, offers a better resolution. It is expected that with the smaller radiometer IFOV, better endmembers could be identified, and thus it would increase the accuracy and reliability of the LSMA model.

If surface snow melting is not the objective of a study, other SSMIS channels with different frequencies and polarization would also benefit from using the unmixing model. Different polarizations may have different sensitivity to the detection target. The higher frequencies of SSMIS, such as 91 GHz, offer much better spatial resolutions. Therefore, using these frequencies may also increase the chance to find better endmembers.

V. Conclusion

The large pixel size of passive microwave images can cause large uncertainty in monitoring surface melt over Antarctic Ice Margins. The use of higher resolution observation can reduce uncertainty in melt maps. This article proposes an LSMA of the T_b time series to calculate the subpixel MI that is comparable with higher resolution products. We found that the map of Boolean MI derived from the SSMIS images

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can overestimate surface melt by 50% in Larsen and Wilkins Ice Shelves. With the treatment of LSMA, the overestimation of surface melt was reduced, and the uncertainty was also reduced.

The second finding from our analysis was that the unmixing method would reduce the MI estimate by a certain amount (e.g., 40%–45% in our study area) regardless of the spatial resolution of the image used. As the partial melting problem causes overestimation of MI, the reduction of MI from the unmixing analysis was correct and necessary. Therefore, we recommend such an unmixing approach to be always applied to surface melt calculation from satellite radiometer data.

The third finding is that the melt fraction is inversely related to the elevation in a log-linear form. The log-linear regression equation between melt fraction and elevation suggests that the partial melting phenomenon at elevation below about 1000 m could be largely controlled by topography, especially near the grounding lines. Therefore, the detection error of surface melting could be a severe problem near the Antarctic ice shelf grounding lines without the unmixing model. On the other hand, however, the topographic complexity near the grounding lines will add difficulty to finding good endmembers. Under such a circumstance, it is recommended to use radiometers with higher spatial resolutions to ensure the quality of endmembers.

To illustrate and verify the time series unmixing method, we focused on the study area of the Antarctic Peninsula. With the evidence collected from this area, we will be able to generalize the application of the spatially constrained LSMA to other major ice shelves of Antarctica and refine the trend analysis of Antarctic surface melt with higher confidence and less uncertainty.

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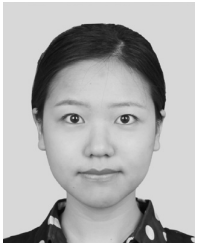
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Dan Tian was born in Changsha, China, in 1990. She received the B.S. degree in geography from Sun Yat-sen University, Guangzhou, China, in 2012, and the M.Res. degree in science of natural hazards from the University of Bristol, Bristol, U.K., in 2013. She is pursuing the Ph.D. degree in geography with Louisiana State University, Baton Rouge, LA, USA.

Since 2018, she has been a Teaching Assistant with the Department of Geography and Anthropology, Louisiana State University. Her research interests include remote sensing, geographic information science, and natural hazards.



Lei Wang (Member, IEEE) received the Ph.D. degree in geography from Texas A&M University, College Station, TX, USA, in 2006.

He is a Professor with the Department of Geography and Anthropology, Louisiana State University, Baton Rouge, LA, USA. His research interests include remote sensing of water and cryosphere, spatio-temporal image analysis, geostatistical models, and flood hazards.

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